

NYC ROAD SAFETY & COLLISION INTELLIGENCE DASHBOARD

Executive Summary

Road traffic accidents remain a significant public safety concern in major cities worldwide. Understanding accident patterns, contributing factors, and high-risk locations is essential for developing effective road safety strategies.

This project analyzes over 238,000 traffic collision records from New York City to identify accident hotspots, injury trends, contributing factors, and vehicle involvement patterns.

The analysis revealed that Brooklyn and Queens experience the highest accident volumes, while driver inattention and distraction remain the leading contributing factors.

Business Problem

Traffic collisions result in:

- Loss of life
- Serious injuries
- Economic costs
- Transportation disruptions

The objective of this project is to:

- Identify accident hotspots
 - Analyze injury patterns
 - Determine major contributing factors
 - Understand vehicle involvement
 - Support data-driven road safety decisions
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Dataset Overview

Dataset Size

- 238,421 Collision Records
- 18 Variables

Key Variables

- Collision ID
 - Date
 - Time
 - Borough
 - Street Name
 - Vehicle Type
 - Contributing Factor
 - Persons Injured
 - Persons Killed
 - Latitude
 - Longitude
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Data Cleaning & Preparation

Data Cleaning Activities

- Replaced missing Borough values
- Replaced missing Street Names
- Replaced missing Vehicle Types
- Replaced missing Contributing Factors
- Standardized categorical values

Feature Engineering

Created:

- Year
- Month Name
- Month Number

These fields enabled trend analysis and dashboard reporting.

Dashboard KPIs

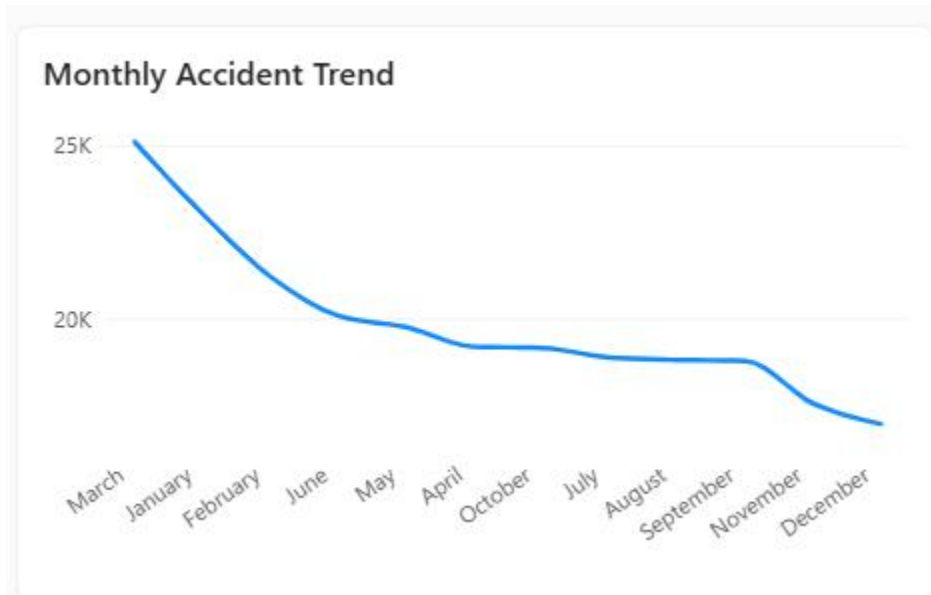
Motorist Injuries	Total Accidents	Cyclist Injuries	Total Injuries	Pedestrian Injuries	Total Fatalities
81K	238K	11K	116K	19K	635

KPIs:

- Total Accidents: 238K
 - Total Injuries: 116K
 - Total Fatalities: 635
 - Pedestrian Injuries : 19k
 - Cyclist Injuries: 11k
 - Motorist Injuries: 81k
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Analysis & Findings

Monthly Accident Trends



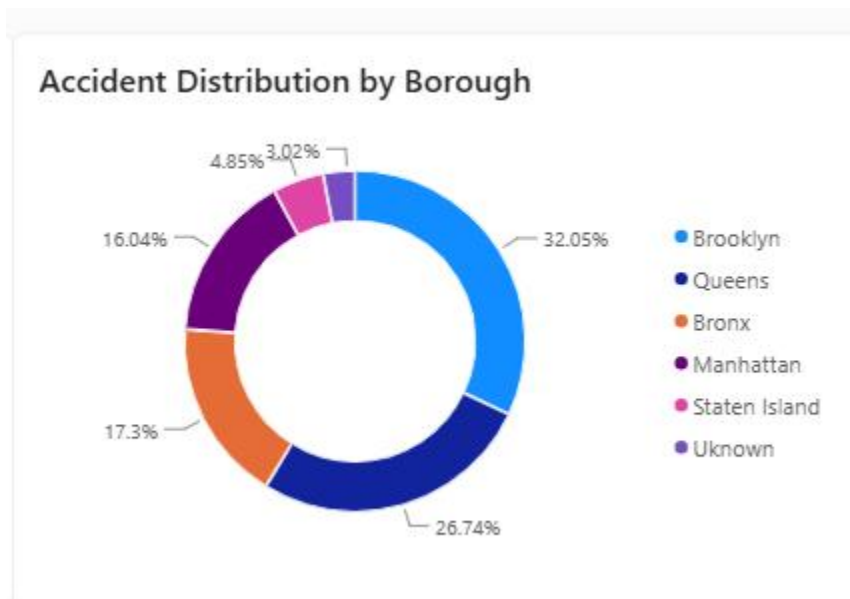
Findings

Traffic collisions fluctuate throughout the year, revealing seasonal and behavioral patterns.

Business Implication

Road safety campaigns can be strategically timed around high-risk periods.

Borough Analysis



Findings

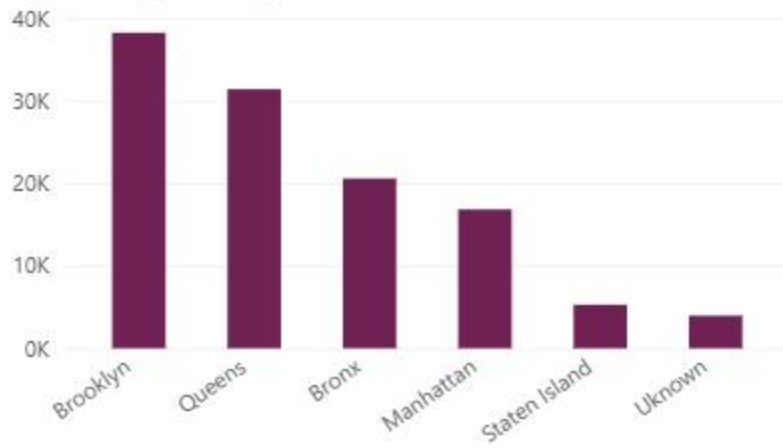
Brooklyn recorded the highest number of accidents, followed closely by Queens. Together, these boroughs account for the majority of collision activity.

Business Implication

Safety interventions should prioritize these locations.

Injury Analysis

Injuries by Borough



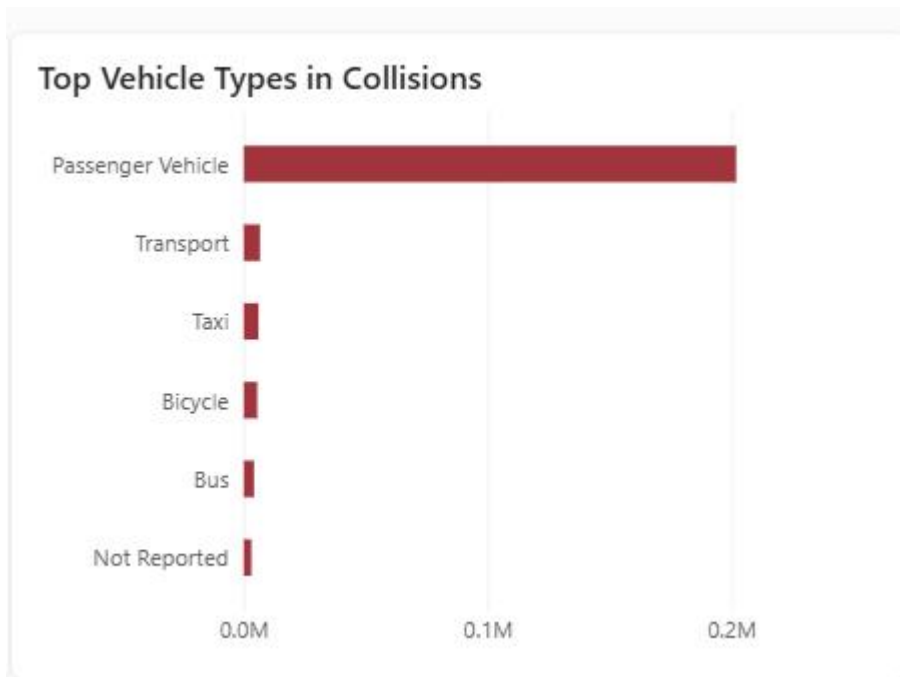
Findings

Brooklyn and Queens also recorded the highest injury counts.

Business Implication

Emergency response resources should be concentrated in these areas.

Vehicle Type Analysis



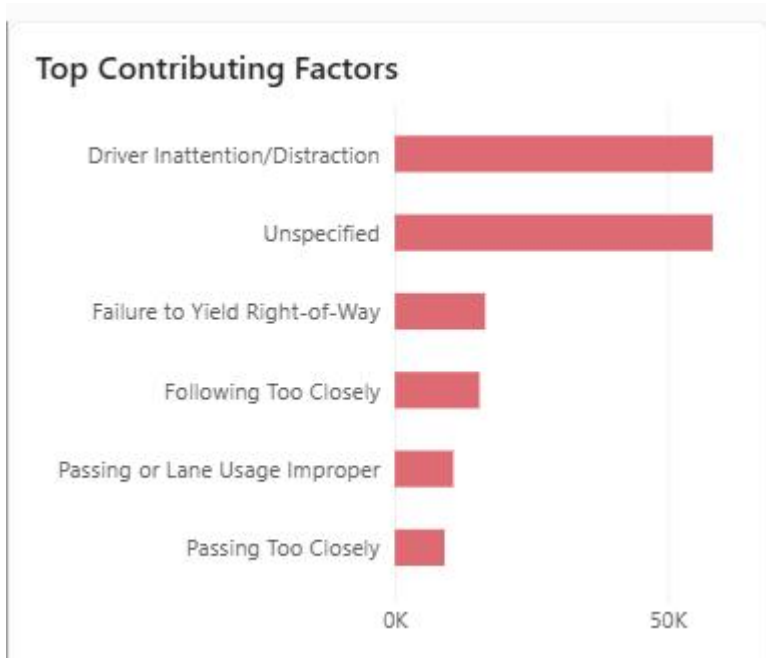
Findings

Passenger vehicles were involved in the majority of reported collisions.

Business Implication

Driver-focused safety programs may have the greatest impact.

Contributing Factors Analysis



Findings

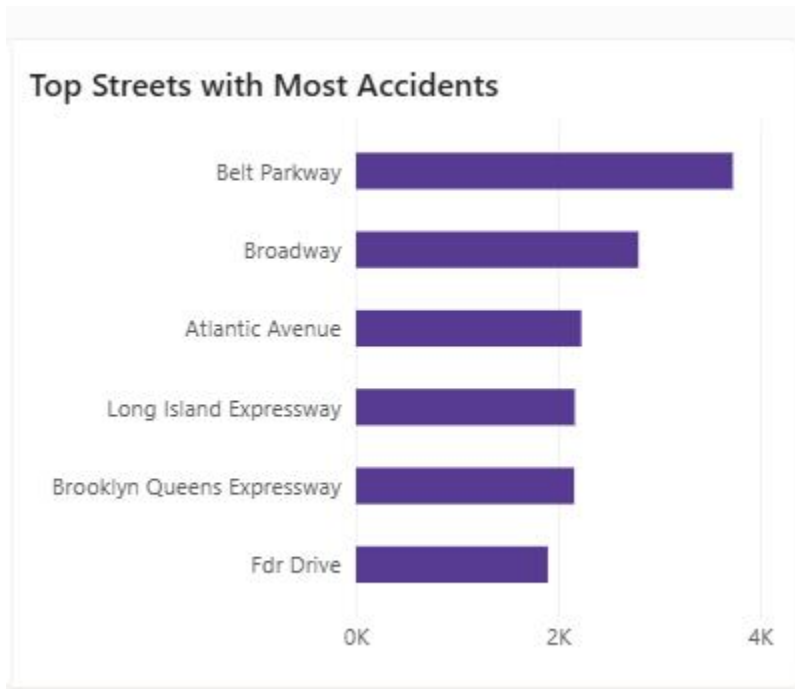
Leading contributing factors include:

- Driver Inattention/Distraction
- Failure to Yield Right-of-Way
- Following Too Closely

Business Implication

Driver behavior remains a major safety challenge.

Streets Analysis



Findings

Several roads consistently experience elevated accident volumes, including:

- Belt Parkway
- Broadway
- Atlantic Avenue
- Long Island Expressway

Business Implication

Infrastructure reviews and safety audits should prioritize these locations.

Key Insights

Insight 1

Over 238,000 collisions were recorded.

Insight 2

More than 116,000 injuries occurred.

Insight 3

Brooklyn and Queens are major collision hotspots.

Insight 4

Driver distraction is the leading contributing factor.

Insight 5

Passenger vehicles dominate collision records.

Insight 6

Specific streets consistently experience elevated accident activity.

Recommendations

Road Safety Recommendations

1. Increase Enforcement in High-Risk Boroughs

Focus enforcement efforts in Brooklyn and Queens.

2. Reduce Distracted Driving

Launch awareness campaigns targeting mobile phone usage and driver attention.

3. Improve High-Risk Roads

Conduct safety audits and infrastructure improvements.

4. Strengthen Traffic Monitoring

Deploy advanced monitoring systems in accident hotspots.

5. Improve Driver Education

Promote defensive driving practices and road safety awareness.

6. Enhance Data-Driven Decision Making

Use predictive analytics to proactively identify emerging collision hotspots.

7. Improve Pedestrian Safety

Expand pedestrian crossings, signage, and safety infrastructure.

8. Support Emergency Response Planning

Allocate emergency services based on collision density and injury trends.

Conclusion

The analysis highlights significant opportunities to improve road safety through targeted interventions, infrastructure upgrades, and behavioral awareness programs.

By focusing resources on high-risk locations and addressing major contributing factors, policymakers can reduce accidents, injuries, and fatalities while improving overall transportation safety.

Full Dashboard Screenshot

